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Anonymization of Online Panel Survey Data: A case study

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Content of the presentation

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- The motivation
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- The problem
- The solution
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  - Anonymization methods and techniques applied
  - Data utility assessment
- Discussion and recommendations

Statistical Disclosure Control for longitudinal data

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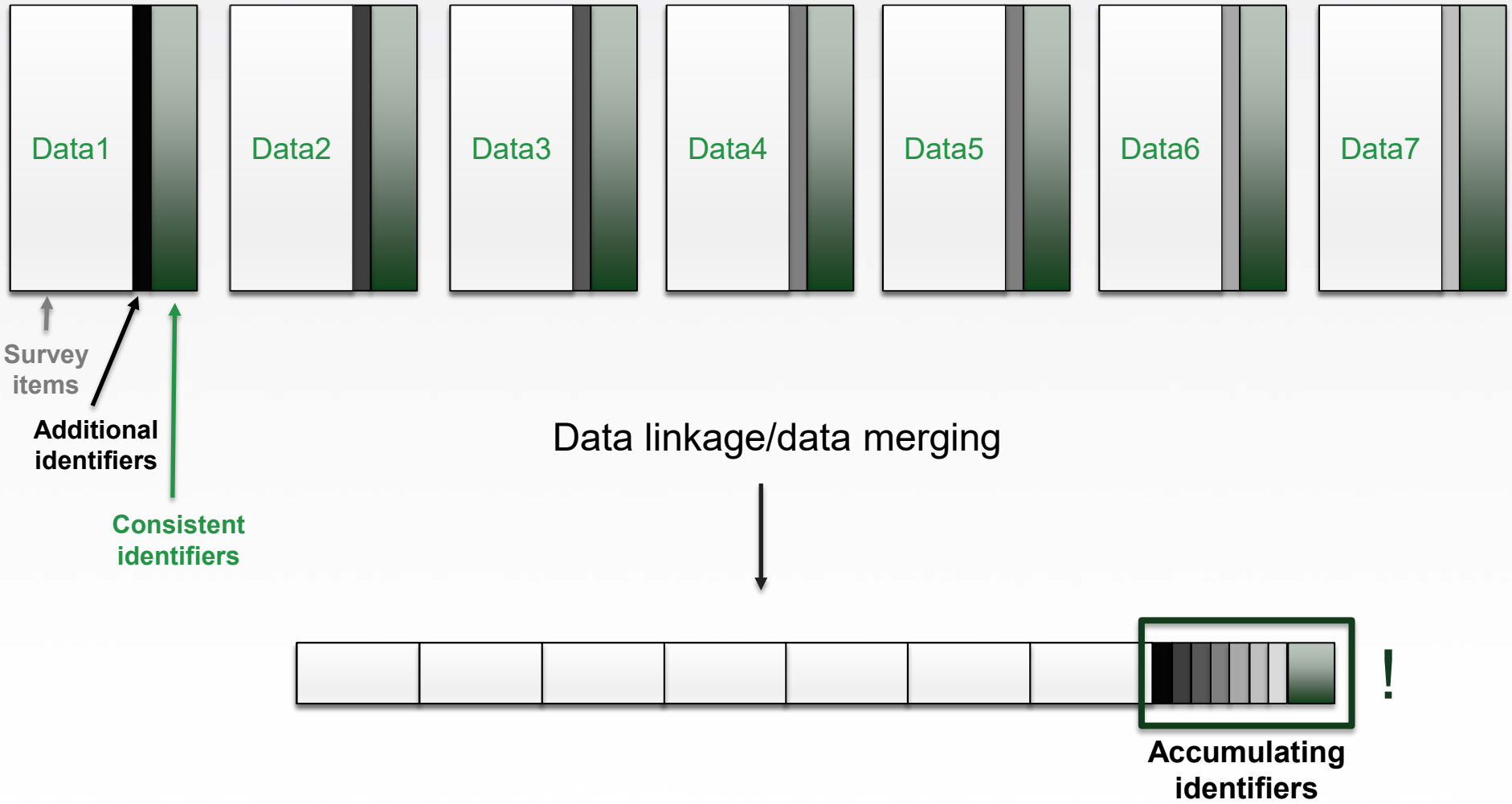
- Anonymization of longitudinal data panel data is a special case
- Traditional SDC approaches might not guarantee protection from identity and attribute disclosure
- Not only socio-demographic and factual information, changes over time can be identifying
- No generally recognised measures for disclosure risk for data collected from the same respondents at different points in time
- Online panel data are not necessarily of longitudinal nature

Dissemination of online panel data

- Online panel is *"a group of selected research participants who have agreed to provide information at specified intervals over an extended period of time"*
- Online panel becoming more and more popular
- Two types: opt-in panels, probability-based panels
- Data can be made available to researchers, students via data repositories/archives, e.g. Australian Data Archive with 7 studies in the catalogue, deposited between 2017-2019
- Confidentiality restrictions should be considered due to the nature of the data

Why do we have to think about SDC of online panel data?



The motivation

- Accumulating demographic and factual information → increased disclosure risk
- Ability of data linkage/merging several waves of online panel data → possible financial gain
- Linking unlinkable data → loss of trust and confidence in the data collector
- Disseminating unsafe data → loss of trust and confidence in the data repository

Methods - data

Title of Life in Australia™ survey	Month and year	Wave	Subsample size
Australian Personas Survey, 2016	December 2016	1	n=2,603
ANU Poll 2017: Housing	March 2017	3	n=2,513
ANU Poll 2017: Job Security	October 2017	10	n=2,270
World Value Survey, 2018	April 2018	14	n=2,106
ANU Poll 2018: Populism	August 2018	19	n=2,220
ANU Poll 2018: Data Governance	October 2018	21	n=2,150
ANU Poll 2018: Population	November 2018	22	n=2,167

Methods - disclosure risk

- Sample frequencies on subset (SUDA2) disclosure risk measure
- Risk is higher if the number of minimal sample uniqueness within an observation is larger
- Risk is higher if the number of variables to determine uniqueness is low
- R package *sdcMicro* includes SUDA2 measure (and others)
- Thinking about disclosure risk of online panel data:

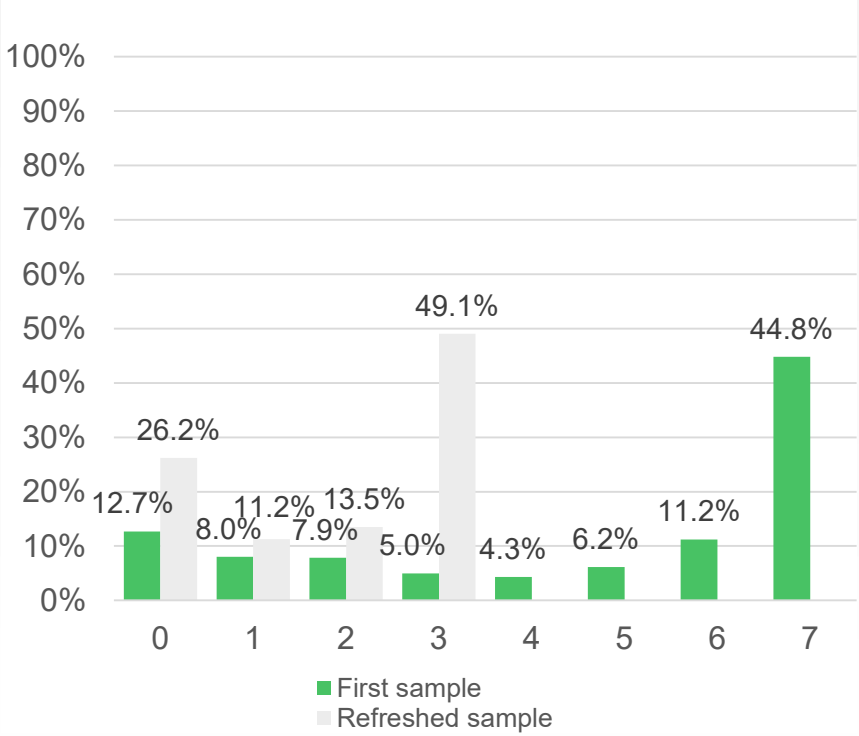
$$\boldsymbol{Disclosure\ risk\ (Online\ panel\ data)} = \ Disclosure\ risk\ (1\ wave) + \\ + Disclosure\ risk\ (Additional\ identifiers) * Record\ Linkage\ Accuracy$$

Methods – data anonymisation

- Data reduction
  - Removing variables
  - Removing cases
  - Global recoding
  - Suppression of values (valid → NA)
- Data perturbation
  - PRAM
  - Micro-aggregation
  - Other
- Synthetic data

The problem 1/3

1.) Respondents in Life in Australia™ participate frequently

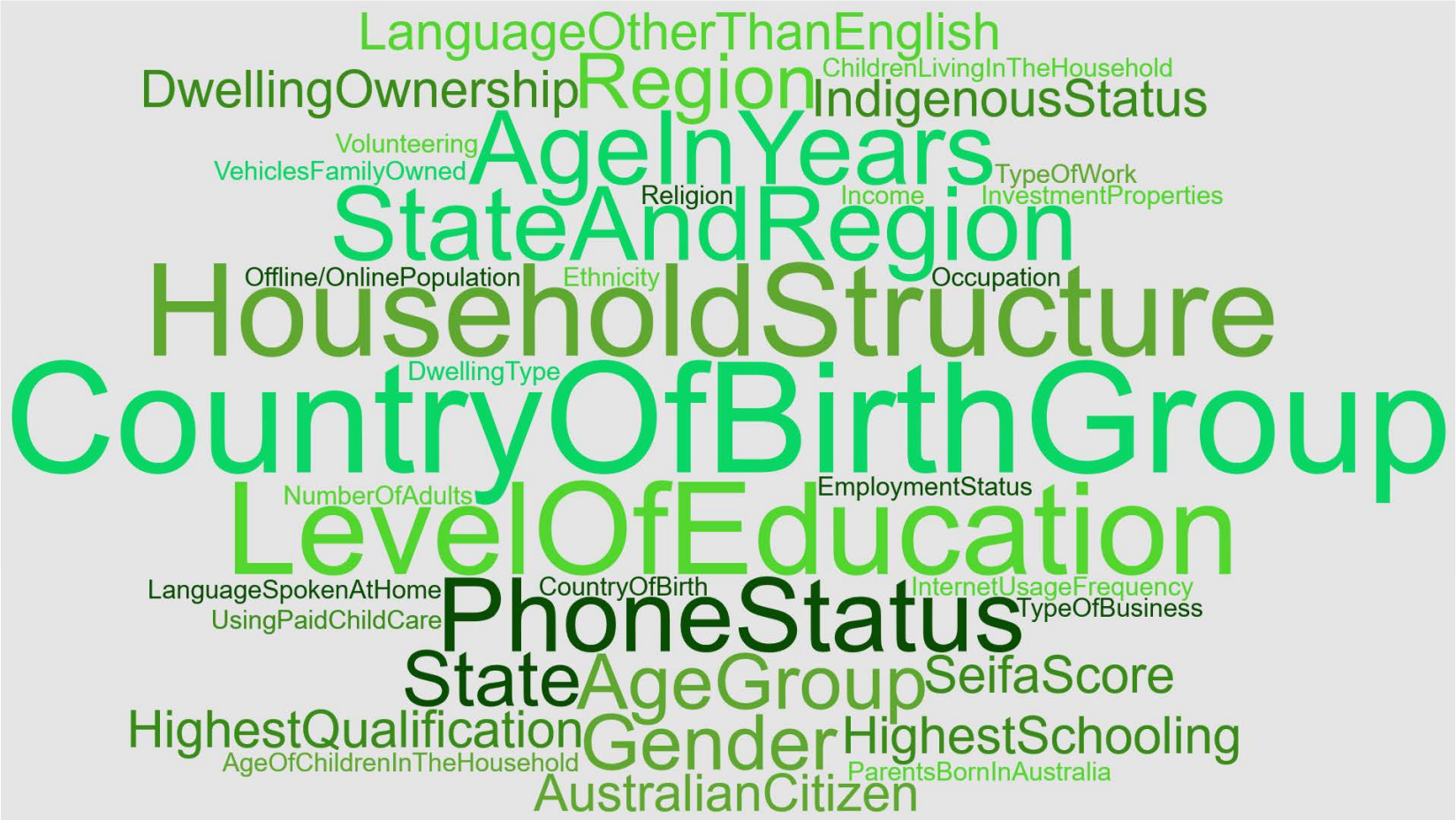


2.) At least half of respondents could be matched in any pair of data files

	W1	W3	W10	W14	W19	W21	W22
W1	78%						
W3	69%	76%					
W10	63%	64%	68%				
W14	59%	59%	60%	63%			
W19	53%	53%	53%	52%	62%		
W21	52%	52%	52%	51%	56%	60%	
W22	52%	52%	52%	51%	56%	56%	60%

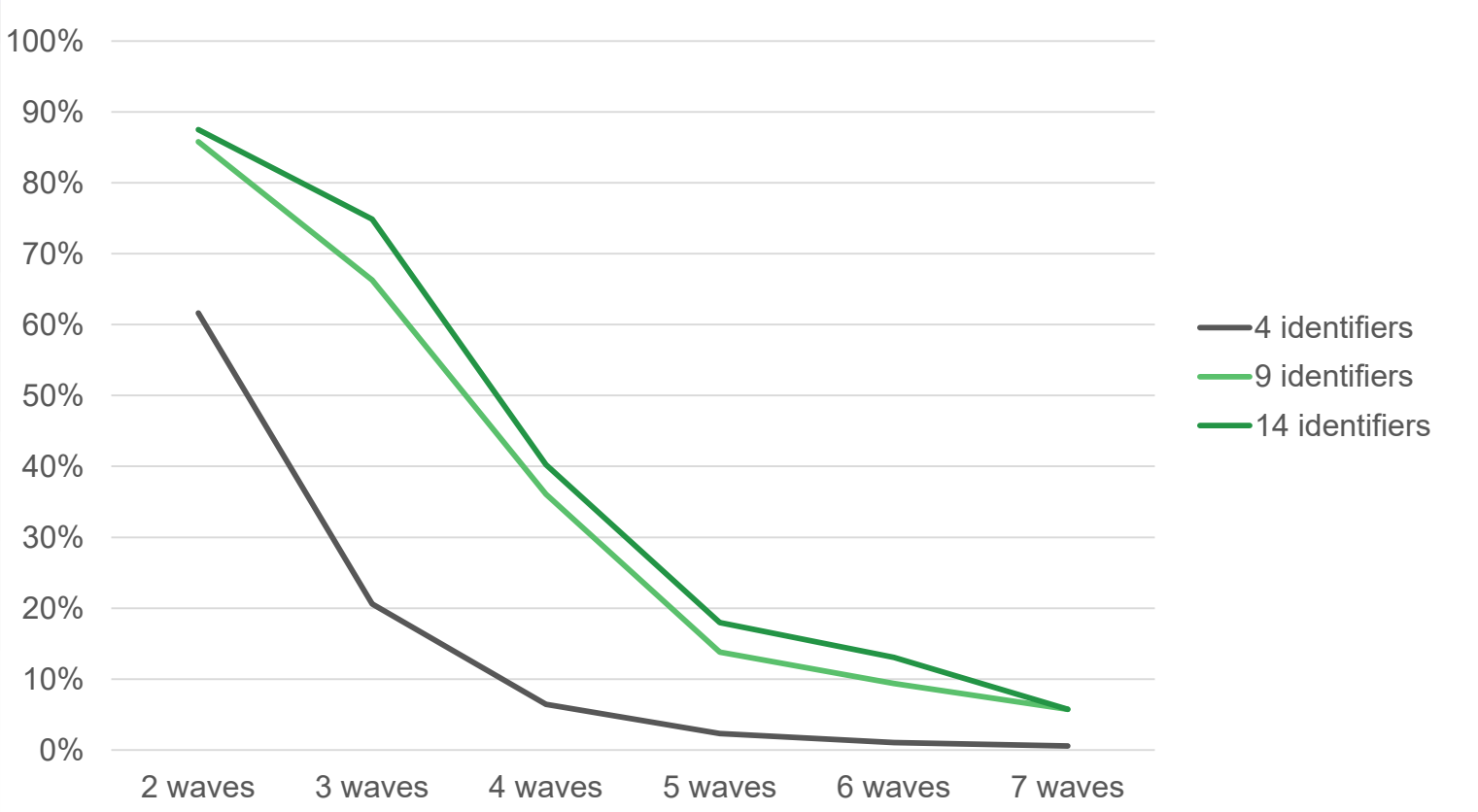
The problem – 2/3

3.) There are a total of about 40 identifying variables available in 7 waves



The problem – 3/3

4.) Data linkage accuracy is high with indirect identifiers



The solution – selection of variables (criteria)

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- importance of variables for statistical analysis, primarily in political science research
- the contribution of variables to the global identification risk and to violating k-anonymity rule
- probability of an intruder having background information about an individual (the least important criterion)

The solution – selection of variables (map)

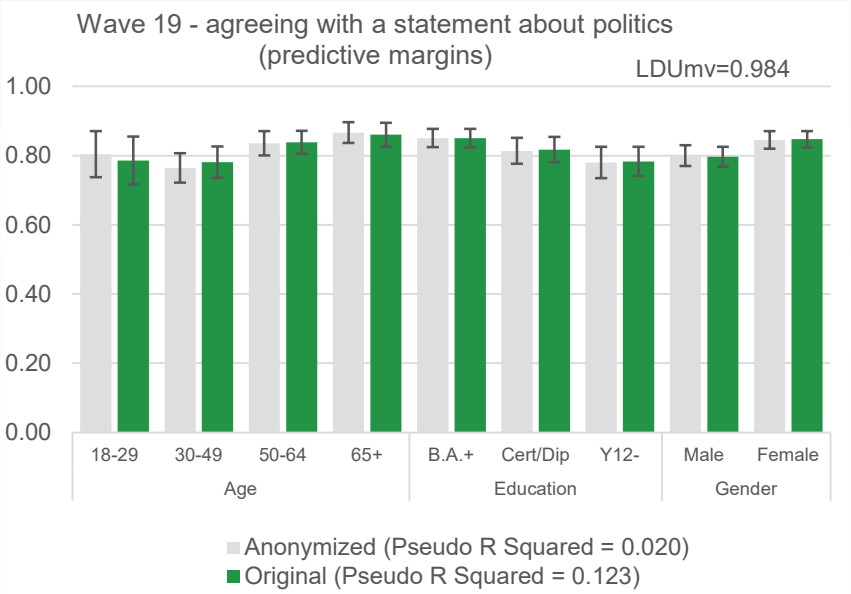
Size of bubble = Probability of having background information

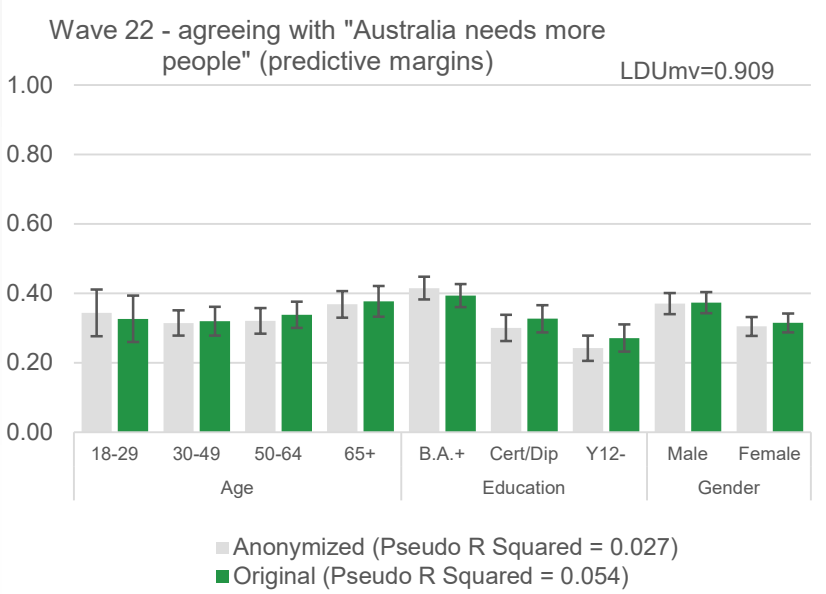
The solution - anonymization methods and techniques

- k-anonymity (k=3) rule
- *gender* (category other recoded to a missing value, local suppression)
- *educatio*n (local suppression only)
- *age group* (reducing the number of age groups from 7 to 4, local suppression)
- *state* (local suppression)
- Other identifiers (variable removal)

The solution - data utility assessment

- Benchmarking approach
- Univariate, bivariate, multivariate level
- An example at the multivariate level:





Discussion

- Data repositories will receive more online panel data for curation and dissemination
- Linking of online panel data is possible, is accurate, increases disclosure risk → SDC is required
- Balance between risk and utility
- Thinking about 5 safes model
- Data repositories can offer data with different levels of SDC under different access conditions

Recommendations

- Data disseminators should think about SDC for data collected from the same respondents at different points in time to earn/keep trust *(data repository perspective)*
- Importance of variables for data analysis and an adequate level of data utility should be considered *(user perspective)*
- Development of disclosure measures for longitudinal and online panel data *(theoretical perspective)*

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Thank you for your attention!

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